

第二章 玻尔模型

光谱

$$\text{氢原子光谱: } \sigma = \frac{1}{\lambda} = R_H \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$

$$m = 1, 2, \dots \quad n = m+1, m+2, \dots$$

玻尔模型

$$1. \quad m \frac{v^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$$

$$E = \frac{1}{2} m v^2 = \frac{e^2}{8\pi\epsilon_0 r} = -\frac{1}{2} \frac{e^2}{4\pi\epsilon_0 r}$$

$$v = \frac{v}{2\pi r} = \frac{e}{2\pi r} \quad \left| \quad \frac{1}{4\pi\epsilon_0 m r}$$

$$\text{辐射条件: } h\nu = \frac{hc}{\lambda} = E_n - E_m \quad \frac{1}{\lambda} = R_H \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$

$$\rightarrow E_n = -\frac{hcR_H}{n^2} \quad \text{得 } R_H = \frac{e^2}{8\pi\epsilon_0 hc} \frac{1}{n^2}$$

2. 对应原理

微观 $\xrightarrow{\text{极限}}$ 宏观/经典 规律一致

$$\nu = \frac{c}{\lambda} = c\sigma = cR_H \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$

取 $n \gg 1, n = m+1$

$$\nu = cR_H \frac{(m+1)^2 - m^2}{m^2(m+1)^2} \approx \frac{2cR_H}{m^3} \quad (\text{微观极限})$$

$$= \frac{e}{2\pi r} \left| \frac{1}{4\pi\epsilon_0 m r} \quad (\text{经典}) \right.$$

$$\Rightarrow r_n = \left(\frac{e}{4\pi\epsilon_0 R_H} \right)^2 \frac{1}{4\pi\epsilon_0 m} n^2 = \frac{e^2}{8\pi\epsilon_0 hc R_H} n^2$$

$$\hookrightarrow R_H = \frac{me^4}{8\epsilon_0^2 h^2 c}$$

$$\hookrightarrow r_n = \frac{\epsilon_0 h^2}{\pi m e^2} n^2 = \frac{4\pi\epsilon_0 \hbar^2}{m e^2} n^2$$

$$E = -\frac{me^4}{8\epsilon_0^2 h^2 n^2} = -\frac{me^4}{(4\pi\epsilon_0)^2 \hbar^2} \cdot \frac{1}{n^2}$$

$$L = e \left| \frac{m}{4\pi\epsilon_0} \right| \frac{4\pi\epsilon_0 \hbar^2}{m e^2} n^2 = n\hbar \quad \text{角动量量子化}$$

3. 数值计算法

1) $r_n = \boxed{r_1} \cdot n^2$ 其中 r_1 为第一玻尔半径

2) $E_n = -\frac{1}{2} m c^2 \cdot \left(\boxed{\frac{e^2}{4\pi\epsilon_0\hbar c}} \right)^2 \frac{1}{n^2} \rightarrow \alpha \approx \frac{1}{137}$ 精细结构常数

$E_1 = \frac{1}{2} m (\boxed{\alpha c})^2 = -13.6 \text{ eV}$
 (基态为0) $\rightarrow v_1 = \frac{c}{137}$ 第一玻尔速度

3) $R = \frac{E_1}{\hbar c}$

4) $r_n E_n = -\frac{1}{2} \alpha \hbar c$

4.

1) 里德伯常数的修正.

$$R_H = \frac{2\pi^2 e^4 m_e}{(4\pi\epsilon_0)^2 \hbar^3 c}$$

考虑了原子核的运动后

$$m_e \rightarrow \mu = \frac{m_e m}{m_e + m}$$

$$R_H' = R_H \frac{\mu}{m_e} = \frac{m}{m_e + m} R_H$$

2) 类氢光谱

$$R_H z = R_H z^2$$

$$\lambda = R_H z^2 \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$

0. 奇异原子

1. 里德伯原子

n 很大: 寿命长, 半径大, 电离能小

2. 奇异原子的体系

$$\text{电子} \rightarrow \mu^-, \pi^-, K^-, \bar{p}, \bar{z}^-$$

$$m_e \rightarrow \mu = \frac{m M}{m + M}$$

$$E_n = - \frac{\mu}{m_e} z^2 \frac{E_0}{n^2}$$

$$r_n = \frac{m_e}{\mu} \frac{n^2}{z} a_0$$

§9 弗克瓦-赫兹实验

3. 相对论修正、玻尔-索末菲模型

$$m = \frac{m_0}{1 - \beta^2}$$

$$E = mc^2 = E_0 - \frac{Ze^2}{4\pi\epsilon_0 r}$$

$$m = \frac{4\pi\epsilon_0 \hbar^2}{mZe^2} n^2$$

$$\hookrightarrow E = (m - m_0)c^2 - \frac{Ze^2}{n^2} m$$

$$= -\frac{1}{2} m \frac{Ze^2}{n^2} c^2 \Rightarrow 2(m - m_0) = \frac{mZe^2}{n^2}$$

$$E = -\frac{m_0 c^2}{2} \left(\frac{Ze}{n} \right)^2 \left[1 + \frac{1}{2} \left(\frac{Ze}{n} \right)^2 \right]$$

碱金属原子的光谱：

$$\sigma = \frac{1}{\lambda} = R_H \left[\frac{1}{n^2} - \frac{1}{n'^2} \right]$$

主线系 : $nP \rightarrow 2S$

锐线系 (第一辅线系) : $nS \rightarrow 2P$

漫线系 (第二辅线系) : $nD \rightarrow 2P$ 3条

基线系 : $nF \rightarrow 3D$ 3条

修正：

① 轨道贯穿效应

② 原子的极化，

